

Research on Power Frequency Waveform Classification for Transmission Lines Based on a Cascaded XGBoost Model

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Abstract—Accurate classification of power frequency waveforms is critical for transmission line fault location. While deep learning models (LSTM, CNN) show promise on idealized data, their practical application is constrained by three major challenges: dependency on simulated data, poor performance on imbalanced real-world datasets, and high computational costs requiring GPU resources. To address these limitations, this paper proposes a cascaded XGBoost model specifically designed for real-grid environments. Our approach decomposes the seven-class waveform classification task into a sequence of binary decisions with tailored feature engineering, enabling effective learning from actual grid data with inherent imbalance. Experimental results demonstrate superior performance over conventional deep learning models, achieving 94.1% overall accuracy, 95.0% macro-average F1-score, and 94.1% weighted-average F1-score while completing training approximately 109 times faster than the LSTM model and 19 times faster than the CNN model. The model's computational efficiency and reliability on CPU-only systems make it particularly suitable for resource-constrained fault diagnosis applications.

Keywords—fault location, power frequency waveform classification, cascaded classifier, XGBoost, feature engineering

I. INTRODUCTION

The reliable operation of overhead transmission lines is frequently threatened by fault-induced trips resulting from lightning strikes, bird contamination, foreign objects, external damage, wind-induced deviations, etc[1]. As these lines often traverse remote and inaccessible areas, pinpointing the exact location of a fault for maintenance crews is a time-consuming, challenging, and potentially hazardous task. To address this, the distributed fault location technology has been developed, which

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involves installing monitoring devices on both ends of a line segment to capture power frequency and traveling waves upon a fault occurrence, transmitting the data to a data center[2]. The data center then employs automatic diagnostic algorithms, for which power frequency waveform classification is an indispensable step, to estimate the fault location[3]. However, the effectiveness of these algorithms is often compromised when facing the complex and variable nature of real-world waveforms. Current classification methods are frequently unreliable. This limitation significantly increases the diagnostic burden on grid operators, potentially delays fault response, and heightens the risk of unsuccessful fault location.

The classification of power frequency waveforms is crucial for the accurate fault location in overhead transmission lines, with various methodologies being explored in the literature. For instance, [4] applied a Long Short-Term Memory (LSTM) network to classify fault types on a simulated transmission line, demonstrating the capability of deep learning models on idealized data. [5] further investigated a hybrid Conformer model for this task, noting a performance gap between validation and test sets, highlighting challenges in generalizing from simulated training data. [6] compared several classifiers, including a standard Artificial Neural Network (ANN), on simulated fault data. Research has also confirmed the strong performance of tree-based models, where [7] utilized a single XGBoost classifier in combination with wavelet-packet transform for fault diagnosis. Similarly, [8] employed a monolithic XGBoost model with feature selection, underscoring the algorithm's effectiveness when paired with expert feature engineering. Beyond the model itself, the classification architecture is significant; the work in [9] illustrated that structuring a multi-class problem as a cascade of simpler classifiers can enhance computational efficiency.

Despite these advancements, current methods predominantly rely on simulated data and face practical deployment limitations due to substantial computational demands requiring GPU acceleration. To address these gaps, this paper presents a novel cascaded XGBoost framework trained on real-world grid data. Our method strategically decomposes the seven-class waveform classification problem into a sequence of binary decisions, incorporating targeted feature engineering at each stage. Experimental results show the proposed approach achieves superior performance with an overall accuracy of 94.1%, macro-average F1-score of 95.0%, and weighted-average F1-score of 94.1%, while completing training approximately 109 times faster than the LSTM model and 19 times faster than the CNN model. The framework provides an efficient, interpretable solution suitable for resource-constrained fault diagnosis applications.

II. METHODOLOGY

A. Problem Formulation

Based on the operational requirements for transmission line fault location and the characteristic patterns of power-frequency waveforms, the waveforms can be categorized into seven distinct types (i.e., Type 1 to Type 7): normal steady-state waveform, abnormal waveform, incoming-power waveform, power-off waveform, tripping waveform, inrush waveform, and disturbance waveform, as illustrated in the figure below.

The primary objective of this research is to develop a cascaded XGBoost model that achieves reliable classification of the seven real-world waveform types. The model is designed to be robust to data imbalance and noise, computationally efficient without requiring GPU acceleration, and capable of meeting the real-time demands of fault location systems.

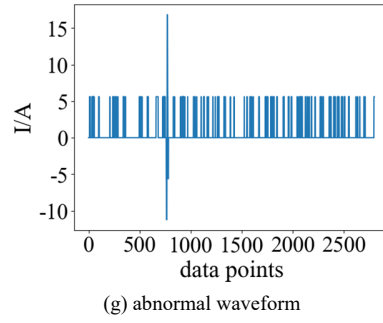
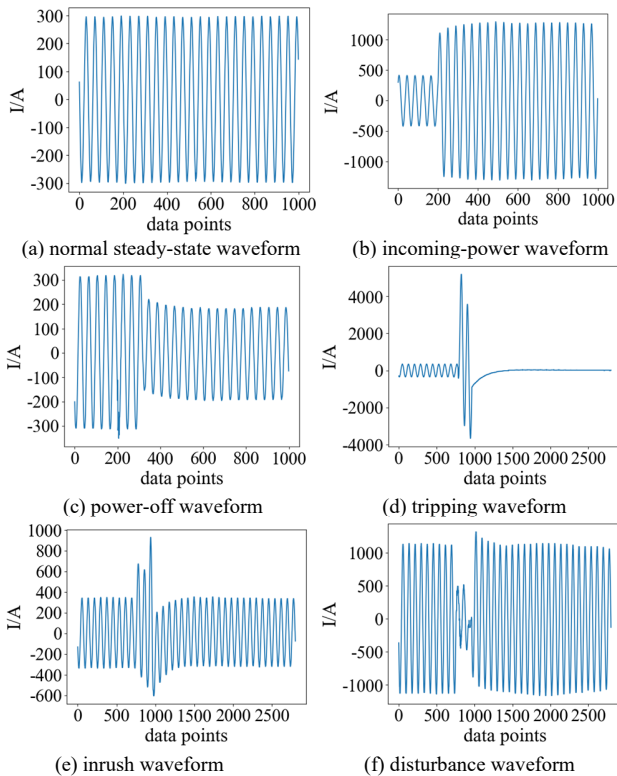


Fig. 1. Power frequency waveforms

B. Methodology: The Cascaded XGBoost Classifier

The cascaded XGBoost model comprises six interconnected XGBoost classifiers (Fig.2), each performing binary classification. When a waveform is processed, it first enters Classifier 1 to distinguish normal vs. abnormal types (e.g., anomaly category in Fig.1(g)). If abnormal, it is directly assigned; if normal, it proceeds sequentially through subsequent classifiers, enabling reliable discrimination of the seven waveform types. This cascaded architecture effectively decomposes the complex task of power frequency waveform classification into a series of simpler XGBoost-based binary decisions, significantly reducing the overall classification difficulty and computational resource requirements.

The cascaded architecture is inherently modular and extensible, designed to accommodate additional waveform categories without reconstructing the core decision logic. If new waveform types are introduced, they can be integrated by inserting new binary classifiers at relevant points in the existing cascade. Each new classifier would focus on distinguishing specific waveform pairs, leveraging the same feature engineering framework and hyperparameter configuration.

The training and application procedure for the cascaded XGBoost model involves five key steps, as illustrated in Fig. 3: waveform data preparation, data annotation, feature selection and calculation, configuration of training hyperparameters, and application of training models.

1) Waveform data preparation:

The waveform data preparation process consists of median filtering and cubic spline interpolation of waveforms. The waveforms used in this study were acquired from real-world fault events on overhead transmission lines, measured by distributed fault location terminals. While this data accurately reflects actual fault scenarios, it often contains high-frequency noise that is detrimental to classification accuracy. Consequently, a median filter is applied to all waveforms to suppress this noise.

The median filter is a classical nonlinear signal processing technique. Its core principle involves sorting the data points within a sliding window (known as the kernel) by their values and then replacing the original data point at the center of the window with the median value[10]. In this study, the kernel size was set to 11. This value provides strong noise suppression for power frequency waveforms while effectively avoiding distortion of the waveform shape. The typical application effect is illustrated in Fig. 4.

Furthermore, the waveforms measured by different distributed fault location terminals may vary in length. Since the XGBoost classifier requires input data of a fixed dimension, all waveforms were standardized to the maximum length present in the dataset using cubic spline interpolation. The cubic spline interpolation algorithm constructs a smooth curve by piecing together a series of simple third-degree polynomial functions. It ensures that the connecting nodes between adjacent polynomial segments are smooth, thereby meeting the smoothness requirements essential for the cascaded XGBoost model[11].

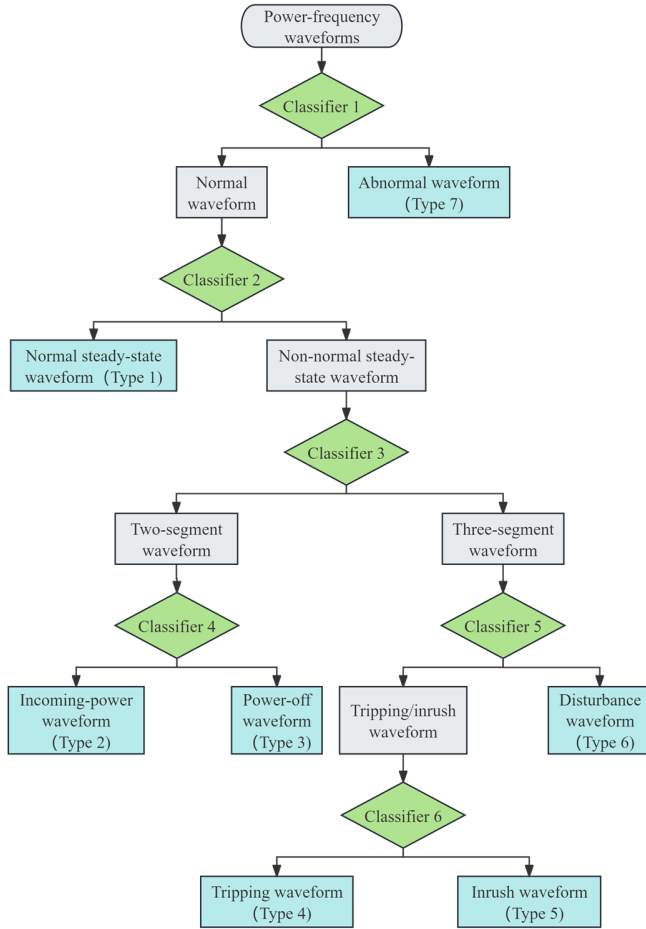


Fig. 2. Cascaded XGBoost model structure

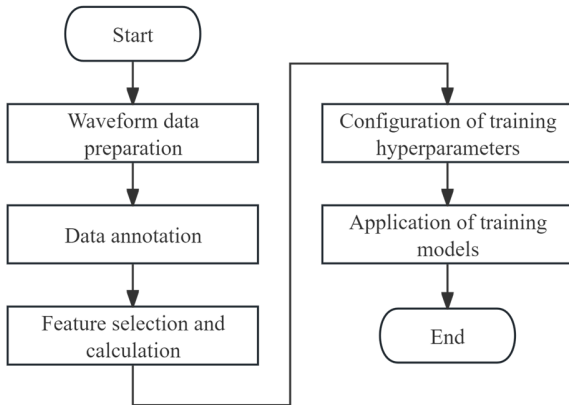


Fig. 3. Training and application procedure for the cascaded XGBoost model

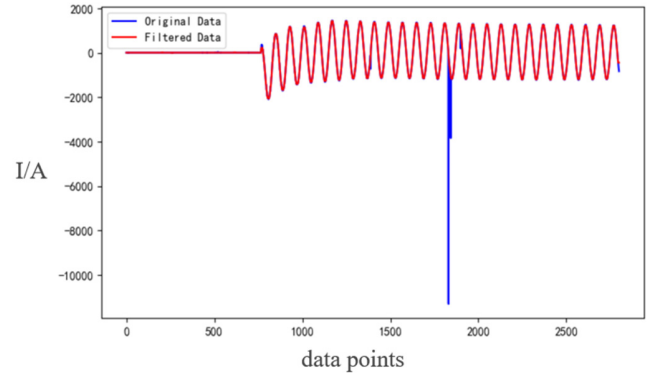


Fig. 4. Typical application effect of the median filter

2) Data annotation:

For the power frequency waveforms that have undergone median filtering and cubic spline interpolation, data annotation is required prior to training. First, all waveform data are labeled according to their true categories (i.e., the seven types shown in Fig. 1). Subsequently, since each XGBoost classifier in this study is designed for binary classification, the labels are adjusted to represent the two specific waveform categories relevant to the corresponding classifier. For example, during the training of Classifier 1, the abnormal waveform type shown in Fig. 1(g) is relabeled as Category 1, while all other waveform types are collectively relabeled as Category 2. This relabeling process is applied correspondingly for each binary XGBoost classifier.

3) Feature selection and calculation:

Following waveform data preparation and annotation, a customized feature set was designed for each binary XGBoost classifier in the cascade, guided by a comprehensive analysis of the physical distinctions between the specific waveform pairs it must differentiate. This ensures that each classifier focuses on the most discriminative characteristics, with all selected features being computed across the entire dataset. The detailed composition of these feature sets for each classifier is elaborated in Section C.

4) Configuration of training hyperparameters:

TABLE I. HYPERPARAMETERS FOR EACH XGBOOST CLASSIFIER

Hyperparameters	Value
n_estimators	1000
max_depth	20
learning_rate	0.01
subsample	0.5
colsample_bytree	0.8
objective	binary:logistic
reg_lambda	1

The hyperparameters for each XGBoost classifier in the cascade were configured to optimize performance for the specific waveform discrimination task at each stage. The complete set of parameters is detailed in Table I. This

configuration employs a large number of weak learners ($n_estimators=1000$) with a low learning rate ($learning_rate=0.01$) for robust convergence, while allowing considerable model complexity ($max_depth=20$) to capture intricate patterns in the waveforms. To prevent overfitting despite this high capacity, strong regularization is applied through feature and instance subsampling ($colsample_bytree=0.8$, $subsample=0.5$) and L2 regularization ($reg_lambda=1$). The binary:logistic objective function is specified for the binary classification nature of each step in the cascade.

5) Application of training models:

Upon completion of the training process, each XGBoost classifier is saved as a model file and reloaded prior to classifying new power frequency waveforms. For inference, the incoming waveforms first undergo the same data preparation and feature calculation procedures applied during training. The processed features are then fed into the cascade of pre-trained XGBoost classifiers, connected sequentially as illustrated in Fig. 2, to generate the final classification result.

C. Feature Design and Extraction

The feature analysis and design for each XGBoost classifier shown in Fig.2 are presented below.

1) Classifier 1:

Classifier 1 is designed to distinguish normal waveforms from abnormal waveforms. The primary distinctions of abnormal waveforms lie in their significantly larger or smaller amplitude and mean value, more intense oscillations, and greater high-frequency components compared to normal waveforms. To capture these differences, the following features are extracted for classification: mean value, standard deviation, amplitude range, high-frequency energy, smoothness, standard deviation and peak of the first and second derivatives, zero-crossing rate, peak count, spectral entropy, and stationarity.

Among these, the amplitude range is defined as the difference between the maximum and minimum values of the waveform; and the zero-crossing rate represents the number of times the waveform crosses zero. Additionally, high-frequency energy is defined as the sum of the absolute magnitudes of frequency components above 150 Hz obtained through Fast Fourier Transform (FFT). The smoothness is calculated as the sum of absolute differences between the original waveform and its median-filtered version. Stationarity is measured as the standard deviation of the mean values computed within sliding windows, where each window covers 100 sampling points of the waveform.

2) Classifier 2:

Classifier 2 is designed to differentiate normal steady-state waveforms from non-normal steady-state waveforms. The key characteristics of non-normal steady-state waveforms include a clear segmentation into two or more distinct sections with significantly different amplitudes, as well as noticeable high-frequency components at the transitions between segments. Additionally, these waveforms generally exhibit reduced mid- and low-frequency components compared to normal steady-state waveforms.

To capture these distinguishing properties, two additional features are incorporated alongside those used in Classifier 1: mid-frequency energy ($50\text{ Hz} < f < 150\text{ Hz}$) and low-frequency energy ($f < 50\text{ Hz}$). These features are calculated similarly to the high-frequency energy metric defined previously. All other features remain consistent with those employed in Classifier 1.

3) Classifier 3:

Classifier 3 is designed to discriminate between two-segment and three-segment waveforms. The primary distinction lies in the number of segments exhibiting significantly different amplitude levels. The feature set for this classifier includes peak statistics, feature differences between the initial and final waveform regions (0%–10% vs. 10%–20%, and 80%–90% vs. 90%–100%), local window characteristics, and wavelet transform-based features.

Peak statistics comprise the standard deviation, mean, amplitude range, sum of amplitudes, and slope of positive and negative peaks, providing a preliminary indication of segment count. Local window features—including mean, standard deviation, and amplitude range within windows of 800 sampling points—are used to capture statistical properties near segment transitions. Wavelet transform features are derived from discrete wavelet transformation using the db4 wavelet with one decomposition level, including the mean and standard deviation of detail and approximation coefficients to highlight abrupt changes in the waveform. Feature differences between the initial and final waveform regions (0%–10% vs. 10%–20%, and 80%–90% vs. 90%–100%) are calculated based on metrics such as mean, standard deviation, amplitude range, and smoothness. These difference features enhance the model's sensitivity to mutations occurring at the beginning or end of the waveform.

4) Classifier 4:

Classifier 4 is designed to differentiate between incoming-power waveforms and power-off waveforms, both of which exhibit a two-segment pattern. The key distinction lies in the relative mean peak amplitudes of the preceding and succeeding segments: in incoming-power waveforms, the mean peak amplitude of the first segment is smaller than that of the second segment, while the opposite holds true for power-off waveforms. To capture this difference, the following features are utilized: the mean peak amplitude in the 0%–20% region versus the 80%–100% region.

5) Classifier 5:

Classifier 5 is designed to discriminate between tripping/inrush waveforms and disturbance waveforms, both of which are three-segment patterns. The primary distinction lies in the trend of the mean peak amplitudes across the three segments: tripping and inrush waveforms exhibit a "low-high-low" pattern, whereas disturbance waveforms follow a "high-low-high" pattern. Due to the variable positioning of transition regions between segments in three-segment waveforms, a simple comparison of mean peak amplitudes—as used in Classifier 4—proves insufficient for reliable classification. Therefore, the following feature set is adopted: peak statistics, local window features, and wavelet transform-based features. The window size and wavelet transform parameters used in Classifier 5 are consistent with those in Classifier 3.

6) Classifier 6:

Classifier 6 is designed to distinguish between tripping waveforms and inrush waveforms. Both waveform types exhibit a three-stage pattern characterized by a "low-high-low" distribution of mean peak amplitudes across the segments. The key distinction lies in the shape and amplitude of the third segment: in tripping waveforms, the third segment decays rapidly to zero or declines gradually following an exponential pattern toward near-zero values; whereas in inrush waveforms, the third segment maintains a near-sinusoidal shape and does not decay to zero. To capture these differences, the following features are utilized: peak statistics, feature differences between the 80%–90% and 90%–100% regions of the waveform, local window features, and wavelet transform-based features. The window size and wavelet transform parameters used in Classifier 6 are consistent with those in Classifier 3.

The existing feature set already incorporates inherent noise robustness through median filtering preprocessing and frequency-domain characteristics. While the current feature engineering demonstrates effective performance under typical noise conditions, the framework supports seamless integration of additional anti-noise features if required for high-noise environments. Potential enhancements could include wavelet denoising coefficients, signal-to-noise ratio (SNR) metrics, or robust statistical features such as median-based spectral features, without altering the core model architecture.

III. EXPERIMENTS AND RESULTS

A. Dataset Description

The power frequency waveform dataset used in this study contains a total of 35441 waveform samples, with the number of samples for Type 1 to Type 7 as follows: 16447, 1003, 289, 897, 2395, 186, 14224. The significant imbalance in sample counts across categories reflects the varying natural occurrence rates of different operating conditions in actual overhead transmission line systems. This authentic distribution avoids the over-idealization that can arise from simulated data, thereby providing a more realistic evaluation scenario. However, it also poses a significant challenge to the classification model, requiring robust performance under highly imbalanced data conditions.

B. Experimental Setup

The experimental hardware environment consisted of an Intel i9-12900K 3.20 GHz CPU, 64GB of DDR4 memory, and an NVIDIA A4000 GPU, with all software running on Windows 11 operating system and Python 3.9. To ensure fair efficiency comparisons, all benchmark models were executed on this identical hardware and software platform. It is noteworthy that the proposed cascaded XGBoost model utilized only CPU resources during both training and inference phases, without employing GPU acceleration.

The selected baseline models include a combination of Long Short-Term Memory network with attention mechanism (LSTM-Attention) and a one-dimensional Convolutional Neural Network (1D-CNN), both representing state-of-the-art deep learning architectures for waveform classification tasks[12]. The comprehensive evaluation metrics for assessing model performance encompass overall classification accuracy,

per-class classification accuracy for each of the seven waveform types, macro-average F1-score, weighted-average F1-score, total training time required for model convergence, and average inference time per waveform sample. All models were trained and tested on identical data partitions to guarantee the fairness and comparability of the comparative results.

The LSTM-Attention model architecture implemented in this study comprises three sequential LSTM layers followed by an attention mechanism and a fully-connected output layer. The model configuration utilizes an input dimension of 1, hidden layer dimension of 128, and output dimension of 7, with a dropout rate of 0.5 applied to prevent overfitting. The dataset was partitioned with 80% allocated for training and 20% reserved for testing. To address the inherent class imbalance in the dataset, the CrossEntropyLoss function was selected as the optimization objective due to its effectiveness in handling imbalanced classification problems. The model was trained using the Adam optimizer with a learning rate of 0.0005, batch size of 32, and underwent training for 100 epochs. The model instance achieving the highest validation accuracy during the training process was selected as the final trained model for evaluation.

For the 1D-CNN model, the architecture comprises three consecutive 1D convolutional layers with increasing filter sizes (64, 128, and 256 respectively), each followed by batch normalization, ReLU activation, and max-pooling operations. The model configuration uses a kernel size of 5 for all convolutional layers, stride of 1, and padding of 2 to maintain temporal dimensions. Following the convolutional blocks, the feature maps are flattened and processed through two fully-connected layers with 128 and 64 units respectively, before the final classification layer with 7 output units corresponding to the waveform categories. A dropout rate of 0.3 is applied after each fully-connected layer to prevent overfitting. The model is trained using the Adam optimizer with a learning rate of 0.001, batch size of 32, and undergoes training for 100 epochs. The dataset follows the same 80-20 training-test split scheme as used for the LSTM-Attention model. CrossEntropyLoss is employed as the objective function to handle class imbalance, and the model achieving the highest validation accuracy during training is selected for final evaluation.

C. Performance Comparison and Analysis

1) Accuracy Performance Comparison:

Regarding classification accuracy, the proposed cascaded XGBoost model demonstrates significantly superior performance compared to both the LSTM-Attention and 1D-CNN models, particularly in its handling of the imbalanced waveform dataset. The comparative results for overall accuracy, macro-average F1-score, and weighted-average F1-score across all models are summarized in Table 2, while the per-class classification accuracy for each waveform type is illustrated in Fig. 5.

TABLE II. COMPARATIVE RESULTS BETWEEN MODELS

Metrics	Classify Models		
	<i>Cascaded XGBoost</i>	<i>LSTM-Attention</i>	<i>1D-CNN</i>
overall accuracy	94.1%	88.6%	87.8%

Metrics	Classify Models		
	<i>Cascaded XGBoost</i>	<i>LSTM-Attention</i>	<i>1D-CNN</i>
macro-average F1-score	95.0%	39.4%	47.6%
weighted-average F1-score	94.1%	86.5%	86.3%

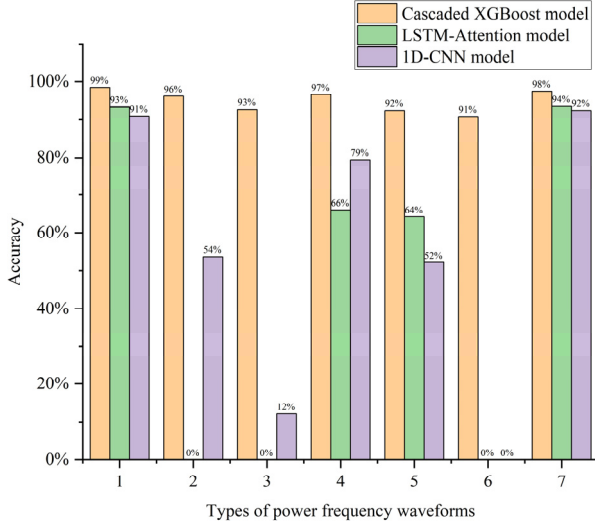


Fig. 5. Per-class classification accuracy for each waveform type

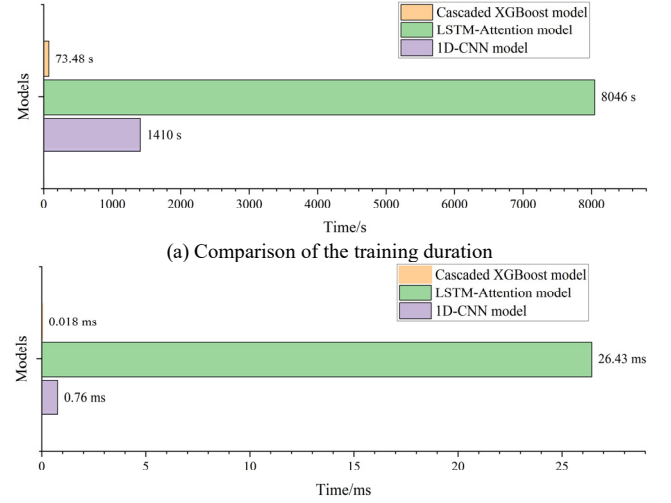
As shown in Table II, the proposed cascaded XGBoost model achieved the highest overall accuracy of 94.1%, significantly outperforming the LSTM-Attention (88.6%) and 1D-CNN (87.8%) models. Furthermore, the results in Fig. 5 demonstrate that the cascaded XGBoost model maintained classification accuracy above 90.9% for all seven power frequency waveform types, while the comparative models only achieved satisfactory performance on majority classes such as normal steady-state and abnormal waveforms, but failed completely to classify several minority waveform types. These results highlight the substantial advantage of the proposed model in both classification accuracy and adaptability to imbalanced datasets when compared to LSTM-Attention and 1D-CNN models.

The significant performance advantage of the proposed cascaded XGBoost model stems from its inherent suitability for imbalanced data. Unlike data-hungry deep learning models (LSTM/CNN), whose gradient-based learning is dominated by majority classes, the proposed model leverages tree-based splits that are naturally robust to class imbalance. Furthermore, the cascaded structure decomposes the complex multi-class problem into simpler binary decisions, each enhanced by targeted feature engineering. This allows the model to effectively learn distinguishing characteristics even from scarce minority-class samples, whereas conventional models often completely fail on these categories.

2) Training and Inference Efficiency Comparison:

In terms of training and inference efficiency, the proposed cascaded XGBoost model demonstrates an advantage by orders of magnitude, significantly reducing computational resource requirements and deployment barriers. Under the

forementioned dataset and hardware environment, the proposed model achieved a total training time of 73.48 seconds, which is approximately 109 times faster than the LSTM-Attention model and 19 times faster than the 1D-CNN model. Furthermore, it achieved an average inference time of 0.018 milliseconds per waveform sample, corresponding to a speed improvement of approximately 1468 times and 42 times over the LSTM-Attention and 1D-CNN models, respectively. A comparison of the training duration and average inference time per waveform sample is illustrated in Fig. 6.



(a) Comparison of the training duration
(b) Comparison of the average inference time per waveform sample
Fig. 6. Comparison of the training duration and average inference time

As indicated in the figure, the training duration for the cascaded XGBoost model represents the sum of the time required to train all six individual classifiers within the cascade. Both the LSTM-Attention and 1D-CNN models were trained for 100 epochs. It is important to note that all reported training and inference times exclusively measure the core model operations and do not include the time spent on waveform data preparation, feature extraction, or other preprocessing steps.

The dramatic efficiency advantage of the cascaded XGBoost model in both training and inference primarily stems from its fundamentally simpler architecture and highly optimized tree-based learning algorithm, as opposed to the computationally intensive gradient-based optimization and complex feature extraction required by deep learning models. Unlike LSTM-Attention and 1D-CNN models, which require massive matrix multiplications, backpropagation through time or layers, and GPU acceleration to be feasible, the proposed model leverages efficient tree learning and a cascaded structure that decomposes the problem into simpler binary tasks. This design eliminates the need for complex feature learning from raw data, and drastically reduces parameter count, resulting in the orders-of-magnitude faster training and instantaneous inference.

IV. CONCLUSION

This paper presents a cascaded XGBoost model for real-world power frequency waveform classification in transmission lines, addressing critical challenges such as data imbalance and computational efficiency. By strategically decomposing the

seven-class problem into a sequence of binary decisions with customized feature engineering, the method achieves 94.1% overall accuracy and 95.0% macro-average F1-score, while training 109 times faster than LSTM and 19 times faster than CNN on CPU-only systems without GPU acceleration. The model's efficiency and reliability make it suitable for resource-constrained applications like edge computing devices.

The research demonstrates how structured traditional machine learning models can surpass deep learning in data-imbalanced industrial scenarios, offering immediate deployability for fault location in conventional transmission lines. For ultra-high voltage (UHV) lines, waveform characteristics may differ significantly, potentially requiring cross-scenario data adaptation to maintain generalization performance. However, the current study focuses on voltage levels from 35kV to 500kV, and any UHV-specific adaptation would be addressed in future work through targeted data collection and fine-tuning within the existing modular framework. Furthermore, future efforts will extend to more fault categories, develop automated feature engineering pipelines, and implement the model on embedded hardware for diverse grid environments.

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